

重庆大学 数据库系统 课程试卷

☒ A卷
☐ B卷

2012 ~2013 学年 第二学期

开课学院： 计算机学院 考试日期： 2013-05-28

考试方式： ☒ 开卷 ☐ 闭卷 ☐ 其他 考试时间： 120 分钟

题 号	一	二	三	四	五	六	七	八	九	总 分
得 分										

注：1. 大标题用四号宋体、小标题及正文用小四号宋体；2. 按 A4 纸缩小打印

NOTES:The exam is open book and notes. Please write your solutions in the spaces provided on the exam. Make sure your solutions are neat and clearly marked. You may use the blank areas and backs of the exam pages for scratch work. Please do not use any additional scratch paper.

Problem 1: (10 points)

As is well known, A DBMS supports concurrent access to data. It can be accessed simultaneously by many distinct processes which are called transactions.

1.Which of the following is not a property of ACID TRANSACTION?

- a) Atomicity
- b) Concurrency
- c) Isolation
- d) Durability

answer is (C B)

2.Please describe the four properties (ACID) of Transaction

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Problem 2: (15 points)

We have the statements of Q1 and Q2:

Q1:
SELECT DISTINCT a
FROM R
WHERE b > 10;

Q2:
SELECT a

FROM R
WHERE b > 10
GROUP BY a;

Which of the following answer is correct?

- (a) Q1 and Q2 produce the same answer.
- (b) The answer to Q1 is always contained in the answer to Q2.
- (c) The answer to Q2 is always contained in the answer to Q1.
- (d) Q1 and Q2 produce different answers.

A

Problem 3. (15 points)

Relation R(a, b, c) currently has the following instance:
{(1; 2; 3), (3; 4; 2), (2; 6; 1)}

We make the following view definitions:

CREATE VIEW V AS
SELECT a*b AS d, c FROM R;

CREATE VIEW W AS
SELECT d, SUM(c) AS e FROM V GROUP BY d;

What is the sum of all the components of all the tuples of the following query?

SELECT AVG(d), e FROM W GROUP BY e;

- (a) 10
- (b) 7
- (c) 23
- (d) 28

A

Problem 4: (15 points).

Consider the following table Games(name, price) and assume that these values already exist in the database: ('game1', 40), ('game2', 50), ('game3', 60). We have the following two transactions:

T1: BEGIN TRANSACTION
S1: UPDATE Games SET price=10 WHERE name='game1'
S2: INSERT INTO Games VALUES ('Game4', 0)
S3: UPDATE Games SET price=30 WHERE name='game1'
COMMIT;

命题人：

罗军

组题人：

罗军

审题人：朱征宇

命题时间：2013-05-15

姓名

学号

年级

专业

学院

T2: BEGIN TRANSACTION
S4: SELECT AVG(price) AS average_price FROM Games
COMMIT;

Above two transactions are hitting the DBMS roughly at the same time.
Suppose T2 must appear to be executed either completely before or completely after T1 , What are the possible values for average_price?

- I. 50
II. 44
III. 30
- a) I only.
b) II only.
c) I & II.
d) I & III.

answer is (A)

Problem 5. (15 points)

Consider the following database schema:
Paintings(artist, work, date, museumname)
Museums(museumname, city, curator)

Consider the following database schema:

We have the statement:

```
SELECT curator
FROM Paintings, Museums
WHERE Paintings.Museumname= Museums.museumname AND
Paintings.artist='LeonardodaVinci' (达。芬奇)
```

Please give the process of the query optimization for the statement above

投影 curator (选择 artist = “LeonardodaVinci” (Paintings 自然连接 Museums))
投影 curator ((投影 museumname (选择 artist = “LeonardodaVinci” (Paintings)) 自然连接 (投影 museumname, curator (Museums)))
Select curator
From (select museumname from Paintings where artist = “LeonardodaVinci”)
natural join (select museumname, curator from Museums)

Problem 6. (15 points)

Let R(A, B, C) satisfy the following functional dependencies (FDs): AB -> C, BC -> A, and AC -> B. The closure of A (i.e., A⁺) is

- a) A
b) AB
c) AC
d) ABC

A

Problem 7. (15 points)

Suppose we have a relation R(A,B,C) and the FD’s
AB-> C, A-> B, B-> C

1. (5 points) What is true about the key(s) for R?

- (a) Only A is a key
(b) Only AB is a key
(c) Only AB and AC are keys.
(d) Only AB, AC, and BC are keys

A

2. (10 points) To decompose R into 3NF

F1={ A-> B } F2={ B-> C }

删除[Unknown]: